

Winston Zhang

+61 493 040 182
winston1c9hv@gmail.com
linkedin.com/in/winston-zhang-89276b208

EXPERIENCE

SYDNEY MOTORSPORTS FORMULA STUDENT TEAM

Sydney, Australia

CAD | Aerodynamic Design | CFD Validation | Composite Manufacturing

Aerodynamics Engineer

July 2022 - December 2023

- Part of the design team for the SM23 Aerodynamics package, mainly focusing on the front wing and undertray.
- **CFD Validation** of the aero package for iterative design and to ensure design goals are met.
- Involved in the **carbon fiber manufacturing** process, including **mold design, manufacture** as well as **carbon fiber layup**.

Aerodynamics Section Lead

December 2023 - February 2025

- **Oversee a team of 10 student engineers** regarding the aerodynamic package's structural design, validation and manufacturing processes. Worked with large, 1000+ part assemblies to ensure adherence to the FSAE regulations.
- Ensure the aerodynamic package functioned as intended through **CFD** on a large computational cluster.
- In charge of the manufacturing for the front and rear wing. Came up with **practical engineering solutions** for various unforeseen issues and designed a structural rib and spar system for the front wing.
- **Managed interpersonal relations** between team members and ensured tasks are allocated and completed on schedule.

Advisor

February 2025 - present

- Working with Sydney Motorsports on validating the aerodynamics package of the vehicle in the University's **wind tunnel** as part of my **graduate thesis**.
- Advise the current aerodynamics team on any manufacturing and validation problems.

DOVETAIL ELECTRIC AVIATION

Sydney, Australia

CAD | CFD Validation | Reverse Engineering

Computational Fluid Dynamics Engineer Intern

September 2023 - January 2024

- Created **parametric SOLIDWORKS model** from 3D scans of the propeller nacelle assembly for easy future modifications.
- **Validate** and alterate the 3D scanned geometry through ANSYS Fluent and comparing the results to the specifications given.
- **Create and optimize ANSYS Fluent** model to minimize resource consumption
- Alterate the cooling duct geometry to minimize drag and maximize mass flow rate.

TECHNICAL SKILLS

• Composite Manufacturing

Prepreg | Hand Layup | Mold Manufacturing

• Computer Aided Design

SOLIDWORKS | Autodesk Inventor

• Numerical Simulation

SOLIDWORKS | ANSYS Suite

• Programming

MATLAB | Python

TECHNICAL PROJECTS

- **Homelab Server** - Acts as a media server and custom cloud storage solution.
- **Robotic Arm** - 6 axis robotic arm with NEMA motors, belt drive and cycloidal gearbox.
- **Custom PC Peripherals** - Designed a 32 gram fingertip mouse and put together a lily58 split ortholinear keyboard and designed a case.
- **UNIX** - Daily drove dual boot windows/linux systems with various distros such as Arch and various Ubuntu variations.

EDUCATION

THE UNIVERSITY OF SYDNEY

Bachelor of Mechanical Engineering Honours (Fluid Mechanics)

Expected Graduation: 2025

2021 - Current

Reference available on request